Exercise 2: Measuring the QSHE with Invasive Voltage Probes

Figure 3 shows the first realization of the quantum spin Hall (QSH) effect, characterized by two counter-propagating edge states of opposite spins, propagating around an insulating two-dimensional bulk. This observation was made by König et al. [Science 318, 766-770 (2007)] in CdTe − HgTe − CdTe quantum wells (QWs), following the theoretical prediction by Bernevig et al. [Science 314, 1757-1761 (2006)]. The band structure of HgTe QWs is described by the (now famous) Bernevig-Hughes-Zhang (BHZ) Hamiltonian:

$$H(\mathbf{k}) = \begin{pmatrix} h(\mathbf{k}) & 0 \\ 0 & h^*(-\mathbf{k}) \end{pmatrix}$$
$$h(\mathbf{k}) = \epsilon(\mathbf{k})I + \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$$

expressed in the basis $|\mathrm{E1}+\rangle, |\mathrm{H1}+\rangle, |\mathrm{E1}-\rangle$, and $|\mathrm{H1}-\rangle$, where $|\mathrm{E1}\pm\rangle$ and $|\mathrm{H1}\pm\rangle$ represent the relevant subbands near the Γ-point, and with $\boldsymbol{\sigma}$ the vector of Pauli matrices. Moreover, $h(\mathbf{k})$ is the 2×2 matrix with

$$\epsilon(\mathbf{k}) = C - D\left(k_x^2 + k_y^2\right)$$
$$\mathbf{d}(\mathbf{k}) = \left(Ak_x, -Ak_y, M(\mathbf{k})\right),$$
$$M(\mathbf{k}) = M - B\left(k_x^2 + k_y^2\right)$$

$H(\mathbf{k})$ thus consists of two copies, $h(\mathbf{k})$ and $h^*(-\mathbf{k})$ of similar forms.

1. [2 Points] The energy term $\epsilon(\mathbf{k})$ is unimportant and can be omitted for our discussion here. Focusing on the other term, what does the band dispersion described by $h(\mathbf{k})$ looks like near $\mathbf{k} = \mathbf{0}$?

   Solution: Ignoring the term $\epsilon(\mathbf{k})$, the Hamiltonian for the subsystem is $h(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$. The energy eigenvalues are given by $E(\mathbf{k}) = \pm|\mathbf{d}(\mathbf{k})|$.

   $$E(\mathbf{k}) = \pm\sqrt{A^2\left(k_x^2 + k_y^2\right) + \left(M - B\left(k_x^2 + k_y^2\right)\right)^2}$$

   Near the Γ-point ($\mathbf{k} = \mathbf{0}$), this simplifies to:

   $$E(\mathbf{k}) \approx \pm\sqrt{A^2k^2 + M^2}$$

   where $k^2 = k_x^2 + k_y^2$. This is the dispersion relation for a **massive 2D Dirac fermion**. The spectrum has a full band gap of size $2|M|$ at $\mathbf{k} = \mathbf{0}$.

2. [1 Point] What fundamental symmetry relates $h^*(-\mathbf{k})$ to $h(\mathbf{k})$? Justify your answer. What can you say about the band dispersion of $h^*(-\mathbf{k})$ near $\mathbf{k} = \mathbf{0}$?

   Solution: The fundamental symmetry that relates $h^*(-\mathbf{k})$ to $h(\mathbf{k})$ is **time-reversal symmetry (TRS)**.

   Justification: The Hamiltonian $H(\mathbf{k})$ is block-diagonal, with $h(\mathbf{k})$ describing one spin sector (e.g., spin-up) and $h^*(-\mathbf{k})$ describing the other (spin-down). The time-reversal operator $\mathcal{T}$ reverses both momentum ($\mathbf{k} \to -\mathbf{k}$) and spin. Therefore, TRS requires that the Hamiltonian for a spin-up electron at momentum $\mathbf{k}$ is related to the Hamiltonian for a spin-down electron at momentum $-\mathbf{k}$. The form $h^*(-\mathbf{k})$ is precisely the time-reversed partner of $h(\mathbf{k})$ for a spin-1/2 system.

   Band Dispersion of $h^*(-\mathbf{k})$: The energy eigenvalues of $h^*(-\mathbf{k})$ are the same as for $h(-\mathbf{k})$. Since the dispersion of $h(\mathbf{k})$ depends only on $k^2 = k_x^2 + k_y^2$, it is isotropic. Thus, the band dispersion of $h^*(-\mathbf{k})$ is identical to that of $h(\mathbf{k})$; it is also a massive Dirac cone with a gap of $2|M|$.

3. [2 Points] Calculate the Chern number $\mathcal{C}$ for $h(\mathbf{k})$ and $h^*(-\mathbf{k})$ for $\mathrm{sgn}(MB) = \pm 1$. For which condition of $\mathrm{sgn}(MB)$ is the system topological?

   Solution: The Chern numbers for the two subsystems depend on the signs of M and B.

   - **Case 1: Trivial insulator** ($\mathrm{sgn}(MB) = -1$) This implies $\mathrm{sgn}(M) = -\mathrm{sgn}(B)$, so $\mathrm{sgn}(M) + \mathrm{sgn}(B) = 0$.
     - $\mathcal{C}_{h(\mathbf{k})} = +\frac{1}{2}[0] = 0$
     - $\mathcal{C}_{h^*(-\mathbf{k})} = -\frac{1}{2}[0] = 0$

- **Case 2: Topological insulator** ($\operatorname{sgn}(MB) = +1$) This implies $\operatorname{sgn}(M) = \operatorname{sgn}(B)$, so $|\operatorname{sgn}(M) + \operatorname{sgn}(B)| = 2$.
  - $\mathcal{C}_{h(\mathbf{k})} = +\frac{1}{2}[\pm 2] = \pm 1$
  - $\mathcal{C}_{h^*(-\mathbf{k})} = -\frac{1}{2}[\pm 2] = \mp 1$

The system is in a **topological** phase when the Chern numbers of the subsystems are non-zero, which occurs for the condition **$\operatorname{sgn}(MB) = +1$**. This corresponds to an inverted band structure.

4. [1 Point] The total Chern number $\mathcal{C}_{\text{total}}$ of $H(\mathbf{k})$ is simply the sum of the Chern numbers of $h(\mathbf{k})$ and $h^*(-\mathbf{k})$. What is $\mathcal{C}_{\text{total}}$? This is an universal result: $\mathcal{C}_{\text{total}} = \ldots$ when time-reversal symmetry is preserved!

Solution: The total Chern number $\mathcal{C}_{\text{total}}$ is the sum of the Chern numbers of the two subsystems.

$$\mathcal{C}_{\text{total}} = \mathcal{C}_{h(\mathbf{k})} + \mathcal{C}_{h^*(-\mathbf{k})}$$

In both the trivial and topological phases, the sum is always zero:

$$\mathcal{C}_{\text{total}} = (\pm 1) + (\mp 1) = 0 \quad \text{or} \quad 0 + 0 = 0$$

A total Chern number of **$\mathcal{C}_{\text{total}} = 0$** is a universal feature of time-reversal invariant topological insulators.

5. [1 Point] Knowing $\mathcal{C}_{\text{total}}$, what would you expect the Hall resistance R_{yx} to be?

Solution: The quantized Hall resistance is given by $R_{yx} = \frac{h}{\mathcal{C}e^2}$. Since the total Chern number $\mathcal{C}_{\text{total}}$ is zero, one would expect the Hall conductance $G_{yx} = \mathcal{C}_{\text{total}} \frac{e^2}{h}$ to be zero. This implies an **infinite or diverging Hall resistance R_{yx}**.

6. [2 Points] $H(\mathbf{k})$ is block-diagonal, hence $h(\mathbf{k})$ and $h^*(-\mathbf{k})$ can be thought of as two decoupled subsystems. How many edge states do you expect for $H(\mathbf{k})$ in the topological regime? Considering that time-reversal symmetry is preserved, what are the propagation directions and spin polarizations of the edge states? Note that the number of edge states of the system cannot be inferred from $\mathcal{C}_{\text{total}}$!

Solution: Since the Hamiltonian is block-diagonal, we can analyze each block separately.

- **Number of Edge States:** In the topological regime ($\operatorname{sgn}(MB) = +1$), the subsystem $h(\mathbf{k})$ has a Chern number $|\mathcal{C}| = 1$. By the bulk-boundary correspondence, it must host **one** chiral edge state. Similarly, the subsystem $h^*(-\mathbf{k})$ also has $|\mathcal{C}| = 1$ and hosts **one** chiral edge state. Therefore, the total system hosts **two** edge states.
- **Propagation and Spin:** Time-reversal symmetry requires that these two edge states are partners. Since TRS reverses both momentum and spin, the two states must be **counter-propagating** and have **opposite spin polarizations**. This pair of states constitutes the "helical" edge transport characteristic of the Quantum Spin Hall (QSH) effect.

7. [3 Points] Use the Landauer-Büttiker formalism to derive the expression for $R_{2\,\mathrm{T}}, R_{\mathrm{yx}}$ and R_{xx} for a QSH insulator. Use the 6-terminal setup shown in the inset of Fig. 3.

Solution: For a QSH insulator with ideal edge channels, we have a spin-up channel propagating clockwise and a spin-down channel propagating counter-clockwise. In the 6-terminal setup, the current at each lead is given by the Landauer-Büttiker formula, assuming full equilibration at the voltage probes: $I_i = \frac{e^2}{h}(2V_i - V_{i-1} - V_{i+1})$.

With current I from lead 1 to 4 ($I_1 = I, I_4 = -I$) and floating voltage probes ($I_2 = I_3 = I_5 = I_6 = 0$), solving the system of equations yields the potentials (with $V_4 = 0$):

- $V_1 = \frac{3h}{2e^2} I$
- $V_2 = V_6 = \frac{h}{e^2} I$
- $V_3 = V_5 = \frac{h}{2e^2} I$

From these potentials, we can derive the resistances:

- **Two-terminal resistance R_{2T}:** $R_{14,14} = \frac{V_1 - V_4}{I} = \frac{V_1}{I} = \frac{3}{2}\frac{h}{e^2}$. The conductance is $G_{2T} = \frac{2}{3}\frac{e^2}{h}$.
- **Hall resistance R_{yx}:** $R_{14,26} = \frac{V_2 - V_6}{I} = 0$. This is consistent with $\mathcal{C}_{total} = 0$.
- **Longitudinal resistance R_{xx}:** $R_{14,23} = \frac{V_2 - V_3}{I} = \frac{(\frac{h}{e^2} - \frac{h}{2e^2})I}{I} = \frac{1}{2}\frac{h}{e^2}$.

This non-zero result for $R_{xx} = \frac{h}{2e^2} \approx 12.9\ \mathrm{k}\Omega$ contradicts the initial prediction of a vanishing voltage drop but matches the experimental data for small devices.

8. [1 Point] Notice, that $R_{2\ \mathrm{T}}$ and R_{xx} depend on the number of voltage terminals present in the device, even when they are floating! What happens to the edge state potentials at the voltage probes?

   Solution: The non-zero longitudinal resistance arises because the voltage probes are **invasive**. Even though no net current flows out of them, they act as reservoirs that connect the counter-propagating edge states. At each probe, electrons from the spin-up and spin-down channels are absorbed and re-emitted, forcing them to equilibrate to a single electrochemical potential. This equilibration process acts as a source of scattering between the two channels, leading to a voltage drop along the sample edge and a finite R_{xx}. The resistance of the QSH state is thus critically dependent on the measurement setup and the number of terminals.

9. [2 Points] Explain the conductance values observed for devices I, II, III, and IV. Any idea why the conductance for device II is not quantized, even though it is in the topological regime?

   Solution: The observed conductance values in Figure 3 can be explained by the topological nature and size of the devices.

   - **Device I:** This device has a quantum well thickness ($d = 5.5$ nm) below the critical value, placing it in the **trivial insulator** regime. At the charge neutrality point, the device behaves as a normal insulator with very high resistance, corresponding to the observed low conductance of $G = 0.01e^2/h$.
   - **Devices III and IV:** These devices are in the **topological regime** ($d = 7.3$ nm) and are small in size. They exhibit a resistance plateau close to $R_{14,23} = h/(2e^2)$, which corresponds to a conductance of $G = 2e^2/h$. This is the quantized value predicted by the Landauer-Büttiker model for ballistic transport along the helical edge states with invasive probes.
   - **Device II:** Although this device is also in the topological regime ($d = 7.3$ nm), it does **not show quantized conductance** ($G \approx 0.3e^2/h$). The reason for this is likely its large size ($20.0 \times 13.3 \mu\mathrm{m}^2$). Over such long distances, inelastic scattering processes can occur, allowing electrons to "shortcut" through the bulk or backscatter between the edges. This breaks the perfect coherence of edge transport, destroying the quantization and leading to a resistance value lower than an ideal insulator but higher than the quantized value.